

quantms-rescoring and multi-engine searching enable deep proteome coverage across quantitative proteomics, immunopeptidomics, and phosphoproteomics

Chengxin Dai^{1,2}, Ralf Gabriels^{3,4}, Robbin Bouwmeester^{3,4}, Jonas Scheid^{5,6,7,8}, Henry Webel⁹, Lennart Martens^{3,4,10,11}, Oliver Kohlbacher¹¹, Mingze Bai¹², Timo Sachsenberg¹³, and Yasset Perez-Riverol^{14,*}

¹State Key Laboratory of Medical Proteomics, Beijing Proteome Research Center, National Center for Protein Sciences (Beijing), Beijing Institute of Lifeomics, 102206, Beijing, China

²International Academy of Phronesis Medicine (Guangdong), 510320, Guangdong, China

³CompOmics, VIB Center for Medical Biotechnology, VIB, Ghent, 9052, Belgium

⁴Department of Biomolecular Medicine, Faculty of Medicine and Health Sciences, Ghent University, Ghent, 9052, Belgium

⁵Department of Peptide-based Immunotherapy, Institute of Immunology, University and University Hospital Tübingen, Tübingen, Germany

⁶Cluster of Excellence iFIT (EXC2180) Image-Guided and Functionally Instructed Tumor Therapies, University of Tübingen, Tübingen, Germany

⁷Quantitative Biology Center (QBiC), University of Tübingen, Tübingen, Germany

⁸Institute for Bioinformatics and Medical Informatics (IBMI), University of Tübingen, Tübingen, Germany

⁹The Novo Nordisk Foundation Center for Biosustainability, Technical University of Denmark, Lyngby, Denmark

¹⁰BioOrganic Mass Spectrometry Laboratory (LSMBO), IPHC UMR 7178, University of Strasbourg, CNRS, Strasbourg, 67000, France

¹¹Infrastructure Nationale de Proteomique ProFI - FR2048, Strasbourg, 67087, France

¹²Chongqing Key Laboratory of Big Data for Bio Intelligence, Chongqing University of Posts and Telecommunications, Chongqing, China

¹³Department of Computer Science, Applied Bioinformatics, University of
Tübingen, Tübingen, Germany

¹⁴European Molecular Biology Laboratory, European Bioinformatics
Institute, Wellcome Genome Campus, Cambridge, United Kingdom

*Corresponding author: yperez@ebi.ac.uk

Abstract

Deep learning methods capable of predicting peptide features in LC-MS/MS —such as retention time and fragmentation intensities— have transformed peptide identification by substantially enhancing the sensitivity and accuracy of established search engines. Here, we present quantms-rescoring an independent component within quantms workflow that enables the integration of major open-source deep learning algorithms like DeepLC, MS²PIP, and AlphaPeptDeep, with quantms-supported search engines MSGF+, SAGE, and Comet. Powered by the Nextflow engine for parallel processing, quantms-rescoring in combination with multiple search engines enables deeper analysis of large proteomics datasets. quantms-rescoring, implements a novel approach to select the best algorithm, model and parameters that fit the data under study. To demonstrate the power of quantms-rescoring, we use the component with four different types of experiments covering label-free quantification (PXD001819, PXD026824), TMT labeling (PDC000127), immunopeptidomics (PXD019643), and phosphoproteomics (PXD026824). Compared to the previous quantms workflow based on a search engine (mainly Comet) and Percolator, this new workflow achieved a 16–22.8% increase in identified spectra, along with the quantification of X% additional phosphorylation peptides and X% phosphosites for the phospho datasets. These improvements demonstrate that integrating multiple search engines with machine learning-derived features not only enhances identification sensitivity but also deepens quantitative insights for downstream biological interpretation. Overall, this workflow offers a reproducible and scalable solution for the reanalysis of public proteomics data, advancing FAIR principles by promoting scientific transparency, accessibility, and data

reuse.

Keywords: Proteomics, Reanalysis, Workflow, Machine learning.

1 Introduction

In recent years, the field of proteomics has experienced rapid growth in the availability of publicly accessible datasets, accompanied by a shift toward studies analyzing larger sample cohorts. As of June 2025, over 40,000 datasets have been submitted to ProteomeXchange (PX) repositories, including a substantial increase in large-scale submissions comprising more than 100 instrument files [1]. Recently, we introduced quantms, an open-source, cloud-based pipeline designed for massively parallel reanalysis of quantitative proteomics datasets [2] to address computational bottlenecks for large-scale quantitative re-analyses. By 2025, quantms supports three major search engines, SAGE [3], MSGF+ [4], and Comet [5], and the combination of them; peptide identifications can be boosted by using Percolator [6]. Different to other proteomics analysis platforms like MaxQuant [7], PeptideShaker [8], or pFind [9], quantms is highly modular and flexible, accommodating a wide range of quantitative proteomics approaches. quantms automatically distributes computations using the Nextflow workflow engine across one or more computing nodes, depending on the number of instrument files and samples [10]. To ensure traceability and reproducibility, the pipeline is built entirely on standardized open file formats [11, 12] and reproducible execution environments using BioContainers Docker and Singularity containers [13], adhering strictly to the FAIR (Findability, Accessibility, Interoperability, and Reusability) principles [13].

Deep learning algorithms that enables to predict peptides features such as retention time; ion mobility; and fragment ion intensities, have transformed the peptide identification landscape, boosting the identification power of existing peptide search engines [14, 15]. Various models have been developed to accurately predict peptide behavior in LC-MS/MS, such as MS²PIP [16], AlphaPeptDeep [17] and DeepLC [18] for fragment ion intensities and retention

time prediction, respectively. These highly accurate predictions enable superior matching of experimental data to theoretical expectations and have reinvigorated rescoring strategies in proteomics. In addition to the deep learning algorithms, a new set of tools and packages has been developed to coordinate, and facilitate the integration of these algorithms with existing search engines and frameworks. MS²Rescore, for example, is a modular Python package that leverages MS²PIP and DeepLC to generate multiple features assessing the similarity between observed and predicted peptide behavior [19].

Here, we introduce quantms-rescoring; an open-source package that enables seamless integration of quantms supported search engines such as SAGE, MSGF+, and Comet and multiple popular deep learning algorithms such as MS²PIP, DeepLC, and AlphaPeptDeep. Different from MS²Rescore, which incorporates the rescoring engine within the package using Percolator or mokapot [20], quantms-rescoring leaves that responsibility to quantms workflow and focuses the efforts on the feature prediction and model fitting to the specific dataset. In addition, the quantms-rescoring package provides a module to compute multiple signal-to-noise spectra features that can be added as features to the rescoring tool in quantms (Percolator). It includes a novel approach to select the best algorithm, model, and parameters that fit the data under study. We benchmarked the new implementation of quantms v1.6.0 using quantms-rescoring for multiple combinations of search engines and deep learning algorithms with different datasets and experimental designs (e.g. different instruments, enzymes, etc) including enhanced pipeline supports in-depth analysis of large-scale public proteomics datasets across diverse experimental designs, including label-free quantification (PXD001819, PXD019643, PXD026824), tandem mass tag-based quantification (PDC000127), immunopeptidomics (PXD019643), and phosphoproteomics (PXD026824) studies. In average, quantms-rescoring achieved a 16–22.8% increase in identified spectra, along with the quantification of X% additional phosphorylation peptides and X% phosphosites for the phospho datasets.

2 Methods

2.1 MS/MS Data and quantms settings

To develop and evaluate the performance of the quantms-rescoring integration with quantms workflow, we selected four publicly available benchmark datasets. Three were obtained from the PRIDE Archive under the identifiers PXD001819 [21], PXD019643 [22], and PXD026824 [23], and a TMT-labeled dataset from the CPTAC data portal under PDC000127 [24]. The PXD001819 standard proteomic dataset contains 48 Sigma UPS1 proteins spiked into a background of yeast cell lysate at nine different concentrations: 0.05, 0.125, 0.25, 0.5, 2.5, 5, 12.5, 25, and 50 fmol/ μ L, that can be used to statistically evaluate label-free approaches for identification and quantification performance due to known spike-in protein of different concentrations. PXD019643 is a large-scale immunopeptide dataset of matched HLA-I and HLA-II ligandomes of benign tissues from HLA Ligand Atlas, which has acquired HLA ligandome data from 29 distinct tissues obtained from 21 individuals, surmounting to 1274 LC-MS/MS runs from 225 mostly paired HLA-I (198) and HLA-II (220) samples. We only selected HLA-II data to evaluate performance of enhanced quantms workflow in immunopeptidomics cases. PXD026824 is a post-translational enrichment dataset for phosphorylated proteins, which is used to evaluate performance on the phosphorylation experiments. PDC000127 is a large-scale TMT10plex-labeled clear cell renal cell carcinoma dataset including 230 samples and 575 LC-MS/MS runs from CPTAC repository, and is used to evaluate performance on TMT experiments and explore potential for reanalysis of large scale public datasets.

Figure 1A illustrates the entire quantms analysis workflow, encompassing label-free quantification (LFQ) and isobaric experiments 1. We evaluated five different quantms workflow settings: (1) Comet, (2) Comet and MSGF+, (3) Comet with quantms-rescoring, (4) Comet and MSGF+ with quantms-rescoring, and (5) Comet, MSGF+, and SAGE with quantms-rescoring to explore multiple search engines and their integration with quantms-rescoring for improved identification and quantification results. SAGE search engine is not considered for

immuno-peptides experiments due to big memory usage. All search parameters for each dataset is described in the original publication (Supplementary File S1). For LFQ, TMT and phosphorylation experiments, FDR filter at 1% threshold was applied at peptide spectrum match (PSM) and protein level. Specifically, the phosphorylation experiment additionally accounted for the rate of local false localization (FLR). For immuno-peptides case, FDR filter at 1% threshold was applied at peptide spectrum match (PSM) and peptide level. The search results from quantms and MaxQuant at the PSM and protein group levels are provided in Supplementary File S1.

@Yasset Comment: @Todo: Dai would be good to add a table with the search parameters for each dataset in the supplementary information and also make public the SDRFs?

2.2 Rescoring Features and Postprocessing

quantms-rescoring computes a broad set of features similar to MS²Rescore by comparing predicted and observed values for (i) retention time from DeepLC and (ii) MS2 fragment-ion intensities from MS²PIP or AlphaPeptDeep, and integrates these features into the quantms workflow. We further extend this with additional spectrum-quality features (signal-to-noise ratio, spectral entropy, fraction of TIC in the top 10 peaks, weighted m/z standard deviation) summarized in Supplementary Table S1, which can be enabled for scenarios such as immuno-peptidomics. To enhance compatibility and performance, we implement explicit strategies for calibration, model selection, preprocessing, and feature export. For retention time (RT), each run is calibrated in DeepLC using a subset of PSMs, and the following features are added per PSM: observed RT, predicted RT, RT difference, as well as best per peptide variants (observed retention time best, predicted retention time best, rt diff best). The DeepLC model selected based on the first run is reused for subsequent runs after per-run calibration [25]. For MS2 intensity features with MS²PIP, spectra are read via OpenMS to ensure consistency with quantms; reporter ions (iTRAQ/TMT) are removed when applicable, spectra are TIC-normalized, and log2-transformed. For a given batch, candidate MS²PIP

models are compared by the average correlation between predicted and observed intensities; the best model is selected if its average correlation is at least 0.4. Model validity is then checked on a calibration subset by removing decoys, ranking PSMs by score, selecting the top 20% (default), and requiring at least 70% of these PSMs to exceed a correlation threshold of 0.6. If these criteria are not met (and not explicitly forced), MS²PIP-based rescoring is skipped to avoid poor fits. For MS2 intensity features with AlphaPeptDeep, we follow an analogous procedure. Predictions are generated from installed or user-provided models, optionally considering neutral-loss fragment ions for PTMs. The same calibration and validity checks are applied (defaults as above). AlphaPeptDeep-derived features quantify agreement between predicted and observed intensities across b/y ions in both log and linear spaces (e.g., Pearson/cosine/dot product, MSE, and distributional summaries). All steps are parallelized at the run level, and users can choose which features to export to the rescoring engine. This design prevents the use of poorly fitting models and filters out pre-trained models that are incompatible with the data, while significantly reducing runtime on large-scale datasets (Figure 1B). Then, the quantms workflow calculates a FDR and posterior error probability (PEP) for each PSM using Percolator. This is performed under three different feature configurations (1) a baseline model using only search engine-derived features, (2) the baseline model plus quantms-rescoring features, and (3) the above combined with SNR features. To merge results from multiple search engines, ConsensusID aggregates PSMs into unified scores based on the calculation of posterior error probabilities (PEP) and chooses the PSM with the highest probability [26]. Final PSM-level q-values are then obtained either directly from Percolator or calculated using OpenMS’ target-decoy approach based on the predicted probabilities. For phosphoproteomics datasets, LuciPHOr2 is employed to assign site-level localization scores and estimate the associated false localization rate using tools from the OpenMS toolkit [27] [28]. All runtime parameters of quantms for different datasets can be found Supplementary file 1.

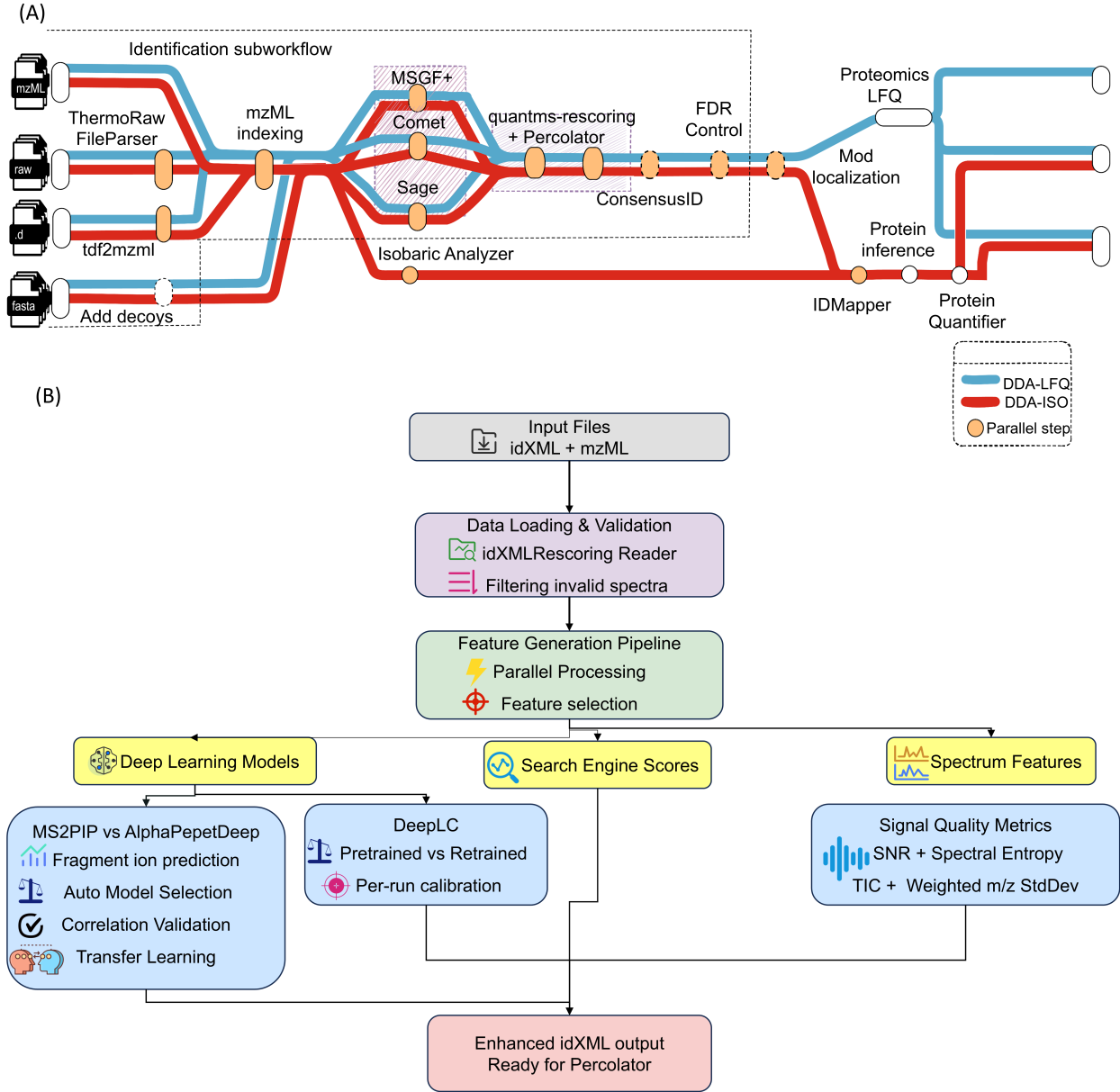


Figure 1: The number of identified spectra as a function of differing FDR levels for different workflow settings. (A) Results for the PXD001819 dataset. (B) Results for the PXD001819 dataset showing protein quantification. Five different workflow settings are shown: the combination of Comet search engine with Percolator rescoring, the combination of Comet with MSGF+

3 Results

3.1 quantms with quantms-rescoring enhances LFQ identification and quantification

To systematically evaluate the performance of the quantms-rescoring-enhanced workflow at both identification and quantification levels, we first analyzed the public UPS1 protein

spike-in dataset PXD001819. For benchmarking the identifications, five different workflows configurations were designed and compared, including (1) Comet with Percolator, (2) two search engines with Percolator, (3) Comet with quantms-rescoring features and Percolator, (4) two search engines with quantms-rescoring features and Percolator, and (5) three search engines with quantms-rescoring features and Percolator to determine whether features from fragment intensity-based and retention time-based predictors improve sensitivity and specificity of identification and quantification.

To compare workflows, significant PSMs were filtered at varying FDR thresholds. As shown in Figure 1-2, using consensus scores from two search engines significantly increased identification rates at a fixed FDR threshold. Specifically, combining Comet and MSGF+ improved identified spectra by 17% over Comet alone, and incorporating MS²PIP and DeepLC features through quantms-rescoring led to an additional 16% increase. quantms achieved a 28% improvement in the number of PSM identifications compared to MaxQuant. At the quantification level, including quantms-rescoring-derived features allowed more low-abundance UPS1 proteins to be quantified (Figure 1B), highlighting the contributions of the integrated workflow. Although MaxQuant reported more UPS proteins in 250amol, 500amol and 2500amol, the proteins that were only quantified in MaxQuant were all quantified by match between runs (MBR) rather than from MS/MS. The main reason for this is that quantms used different MBR quality control. Therefore, we can observe that quantms achieved lower coefficients of variation (CV) than MaxQuant, considering only proteins that are quantified in at least 50% of the replicates (Figure 1C-D). On the other hand, we also found that the variability of proteins quantified were not significantly reduced or increased after enabling quantms scoring.

To better understand the contribution of individual features, we extracted the top 20 feature weights from Percolator (Figure S1). Over half of the top-weighted features were derived from quantms-rescoring. Notably, SpecPearsonNorm had a strong positive weight, indicating that a better correlation between predicted and observed intensities improves

confidence. Conversely, RtDiffBest had a negative weight, suggesting that large deviations in retention time reduce match quality. Interestingly, peptide length also emerged as a significant positive feature after quantms-rescoring integration, likely because peptide length is an important feature as proxy to determine value of other specific features. For instance, for very short peptides, features like pearson correlation could be less valuable and should be ignored.

3.2 Improved identification and quantification in tandem mass tag (TMT) experiments via quantms-rescoring

We next applied the workflow to a large-scale TMT-labeled dataset from the Clinical Proteomic Tumor Analysis Consortium (CPTAC; PDC000127). The integration of multiple search engines and quantms-rescoring led to substantial improvements in both identification and quantification (Figure 3). Compared to Comet with Percolator, multiple search engines consensus identifications with Percolator. The PSM identification rate increased by 3.6%, and 921 proteins were newly quantified compared to the workflow without quantms-rescoring (Figure 2A, 2B).

Among these, 59 proteins had abundance levels within the top 10% (Figure 2C). To assess the biological impact of rescoring, we conducted differential expression analysis using the quantms-rescoring-enhanced workflow. As shown in Figure 2D, 27 newly quantified proteins were significantly differentially expressed. Notably, FOXG1—one of these proteins—has been associated with prognosis in Clear Cell Renal Cell Carcinoma, as reported by Yang et al. [29]. These findings illustrate that quantms incorporating MS²PIP and DeepLC-derived features enhances not only identification sensitivity but also enables deeper biological insights.

We further examined the rescoring contribution of individual features by analyzing the top 20 support vector machine (SVM) weights from Percolator (Figure S2). For Comet-based results, SpecPearsonNorm, DotProdIonYNorm, and RtDiffBest were highly weighted. The interpretations of these weights were consistent with those from the label-free dataset.

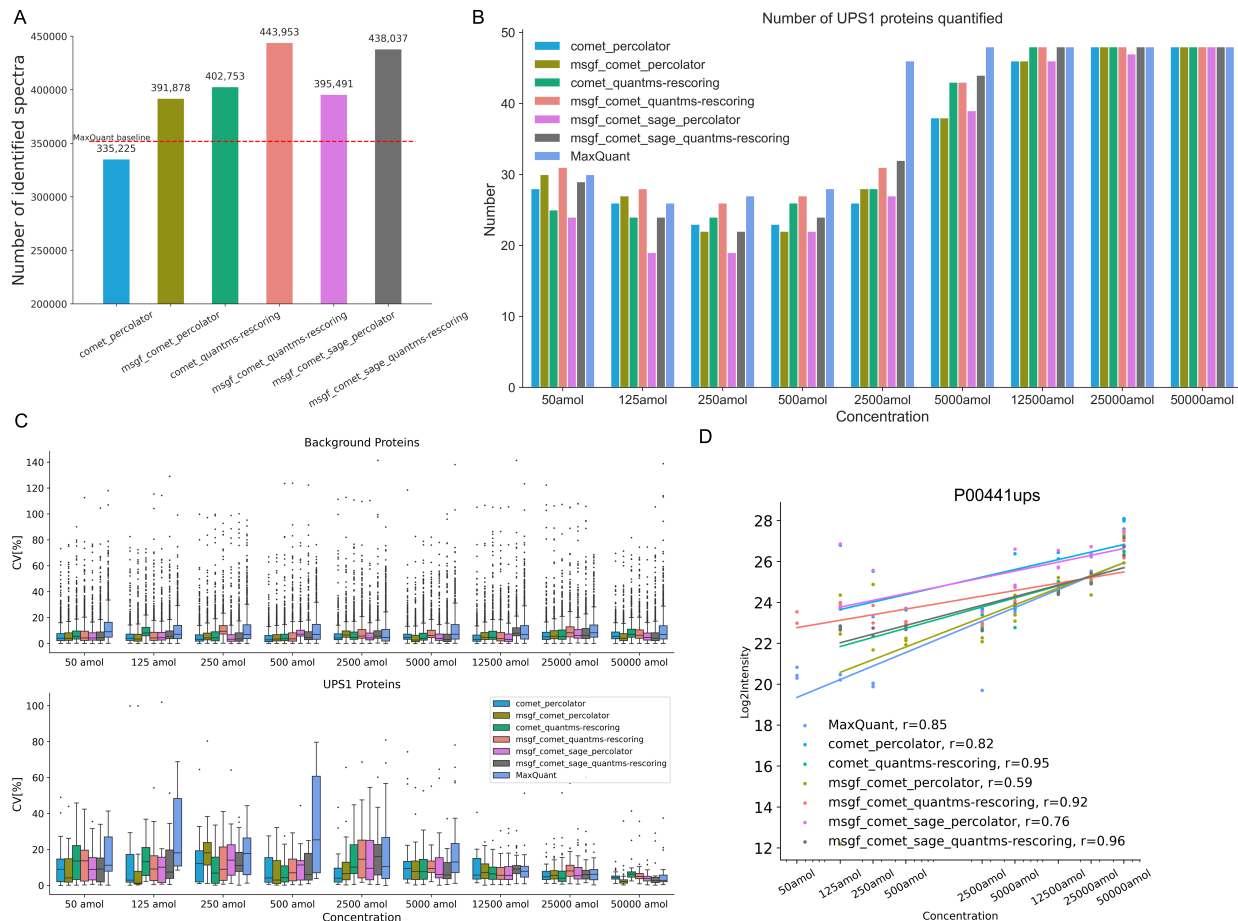


Figure 2: The number of identified spectra as a function of differing FDR levels for different workflow settings. (A) Results for the PXD001819 dataset. (B) Results for the PXD001819 dataset showing protein quantification. Five different workflow settings are shown: the combination of Comet search engine with Percolator rescoring, the combination of Comet with MSGF+, the combination of two search engines with quantms-rescoring features, the combination of two search engines with quantms-rescoring and SNR features, and the combination of three search engines with quantms-rescoring and SNR features. Multiple search engines consensus identification, additional intensity-based and retention time-based features, and SNR features lead to both higher identification rates and high-quality identification. Coefficient of variation of Yeast (C) and UPS1 proteins (D) for the reanalysis of PXD001819 using MaxQuant and quantms with different configurations.

Importantly, peptide length again emerged as a key feature. Similar weight distributions were observed for MSGF+ and SAGE results, confirming the generalizability of these feature effects across different search engines. (Supplemental Figure ??)

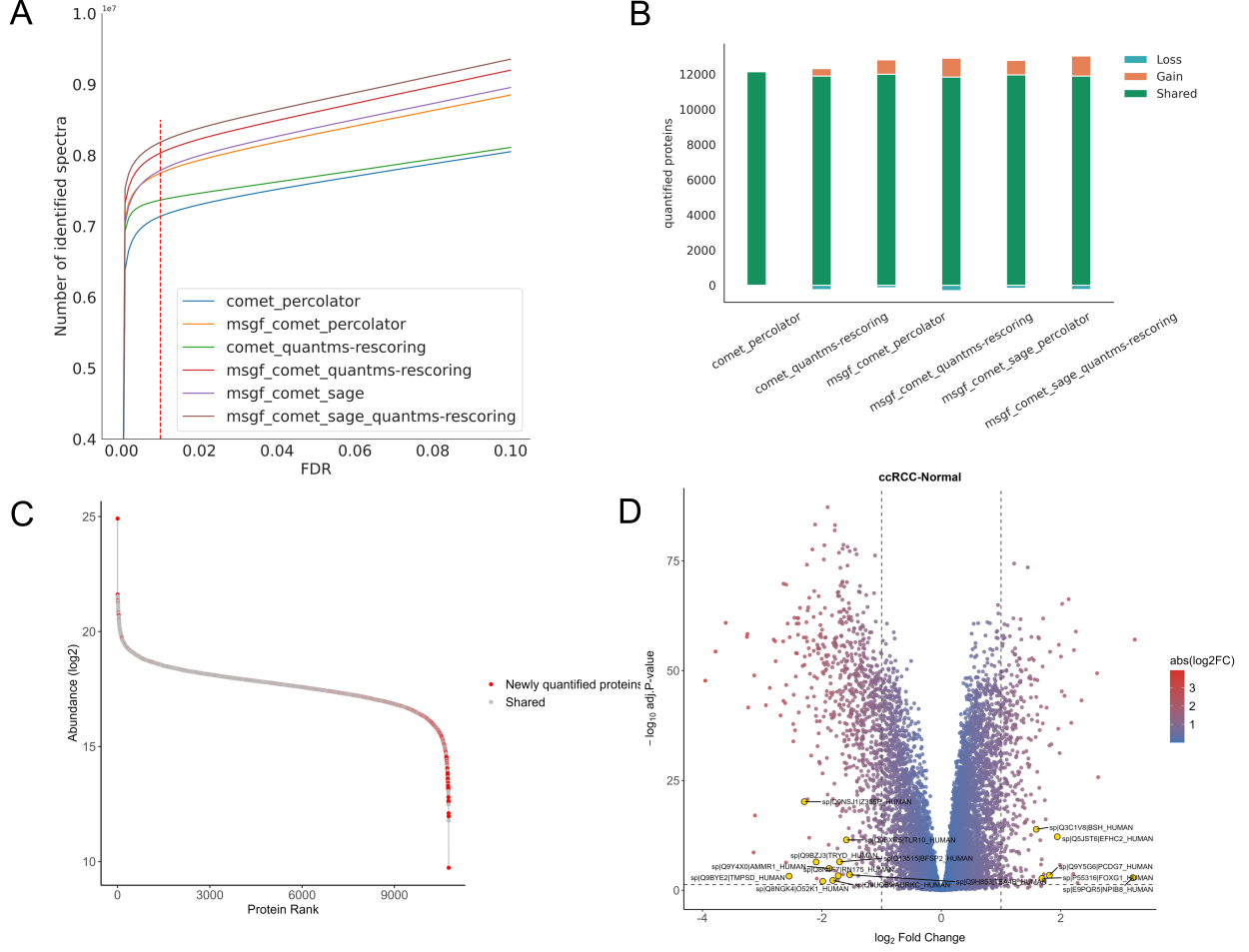


Figure 3: Comparison of identification and quantification results for different workflow settings on PDC000127. (A) The number of identified spectra as a function of differing FDR levels for different workflow settings. (B) The number of quantified proteins. The green part indicates the intersection between a workflow and Percolator-Comet results. (C) The rank of protein abundance from Comet and MSGF+ with quantms-rescoring. The red dots represent proteins quantified only in the Comet and MSGF+ with quantms-rescoring workflow compared to the Comet and MSGF+ workflow without quantms-rescoring. (D) The volcano plot for differential expression analysis from MSstatsTMT. The yellow dots represent proteins quantified only in the Comet and MSGF+ with quantms-rescoring workflow compared to the Comet and MSGF+ workflow without quantms-rescoring.

3.3 quantms with quantms-rescoring improves immunopeptidome identification

The HLA Ligand Atlas [22] (PXD019643) is a comprehensive map of HLA-I and HLA-II-presented peptides from 30 benign tissues, 51 HLA-I alleles, and 86 HLA-II alleles. To further

assess the performance of the quantms workflow using quantms-rescoring, we analyzed the HLA Class II data of the HLA Ligand Atlas. Five workflow configurations were evaluated: (1) Comet with Percolator, (2) Comet with quantms-rescoring, (3) Comet and MSGF+ with Percolator, (4) Comet and MSGF combined with quantms-rescoring features, and (5) Comet and MSGF+ combined quantms-rescoring/SNR features, displayed in Figure 4. With local peptide-level FDR threshold of 1% among the technical replicates per sample, integrating MSGF+ with Comet increased the number of identified peptide by 1.6% compared to using Comet alone. Adding quantms-rescoring features on top of Comet workflow yielded a further 14% improvement compared to MSGF+ with Comet workflow. Enabling the quantms-rescoring combined with MSGF+ and Comet, the peptide identification improved by an additional 5% compared to workflows using only Comet and quantms-rescoring. When considering spectra features such as SNR in Supplementary Table S1, the peptide identification improved by an additional 0.6% compared to workflows using MSGF+, Comet and quantms-rescoring. To demonstrate the application potential of the quantms pipeline for antigen discovery, we show the number of HLA class II peptides predicted as binder (Figure 4B,C and D). Added quantms-rescoring features to the Comet workflow, enabling the identification of 8,867 additional binding. When quantms-rescoring was enabled in combination with MSGF+ and Comet, an additional 2,821 binding were identified. And the length distributions of HLA class II peptide binders predicted are consistent by different workflows configurations. It displays a local maximum occurring at approximately 15-mers (Figure 4E).

In addition, using local PSM-level FDR threshold of 1%, integrating MSGF+ with Comet increased the number of identified spectra by 11.7% compared to using Comet alone. Adding quantms-rescoring features on top of the multi-engine workflow yielded a further 22.8% improvement (Figure S3).

To better understand the impact of quantms-rescoring-derived features on rescoring, we visualized the distributions of decoy PSMs, rejected target PSMs, and accepted target PSMs

using the Pearson correlation coefficient (PCC) and retention time error (Figure S3C,D). The accepted target PSMs were clearly separated from both decoy and rejected targets based solely on PCC values, and exhibited consistently low retention time deviations. The top 20 feature weights from Percolator are displayed in Figure S4. The features of the MSGF+ search engine carry significant weight, such as SpecEValue, which indicates that when only MSGF+ is used, the gain of quantms-rescoring is limited. However, features from quantms-rescoring occupy a larger weight for Comet search engine, including 9 MS²PIP and 2 DeepLC features. This further illustrates that using consensus identifications by multiple search engines helps to coordinate the differences between different search engines. Overall, these results demonstrate the robustness of the quantms workflow in immunopeptidomics, highlighting its ability to leverage machine/deep learning learning-derived features to maximize identification accuracy through multi-engine consensus rescoring.

3.4 Enhanced Phosphopeptide Identification and Localization Using quantms-rescoring

In addition, we investigated the performance of our quantms workflow with quantms-rescoring on post-translational modification experiments (PXD026824). For phosphoproteomics analyses, different workflow settings were evaluated: (1) Comet + Percolator, (2) Comet + quantms-rescoring, (3) Comet + MSGF+ + Percolator, and (4) Comet + MSGF+ + quantms-rescoring features. As shown in Figure 5, combining search engine consensus results with quantms-rescoring features substantially improves Percolator’s ability to discriminate between true and false PSMs. This is evident from the leftward shift in score distributions compared to the Comet-only workflow, indicating improved stringency (Figure 4A). Specifically, incorporating quantms-rescoring features into the two-search-engine consensus increased the number of identified spectra by 24% compared to consensus identification alone. Previous research suggest that phosphorylated peptides may produce -98 Da phosphoric acid neutral losses. Therefore, we further evaluate the impact of considering neutral

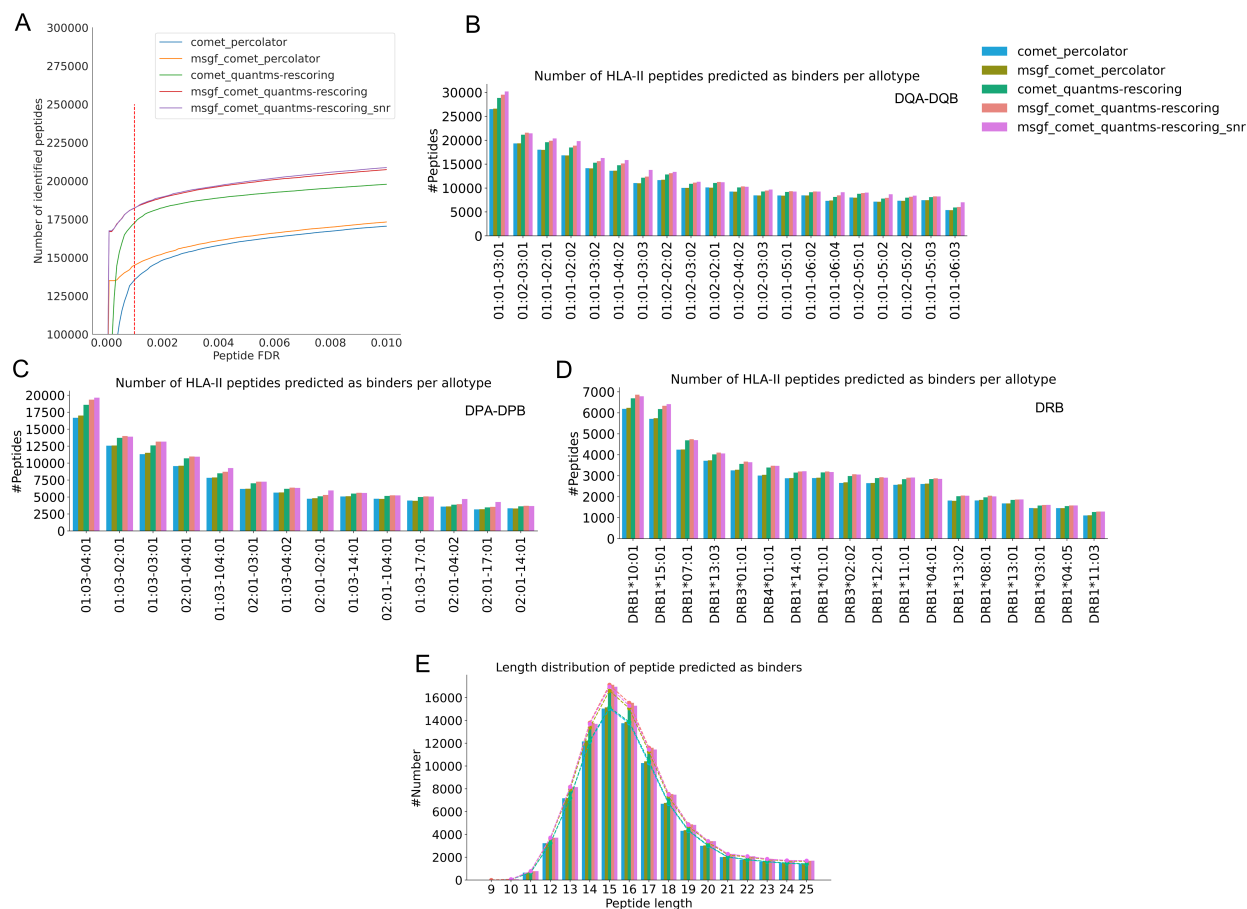


Figure 4: Comparison of immunopeptides identification results for different workflow settings on PXD019643. (A) Line plot showing the number of peptide identified for four workflow settings at different peptide FDR levels. (B, C, D) Global overview of HLA-II predicted binders distributed across HLA molecules. HLA binding prediction was performed with NetMHCIIpan-4.3(% binding rank ≤ 5). (E) Length distribution of identified HLA-II peptides predicted as binders from all samples was analyzed.

loss on re-scoring performance when calculating features. quantms-rescoring with modloss ions increased the number of identified spectra by 3% compared to without modloss ions. The top 20 feature weights from Percolator, displayed in Supplementary Figure ??, highlight the consistent importance of features such as IonYNorm and DotProdIonYNorm, similar to previous datasets.

Given the importance of false localization rate (FLR) control in phosphoproteomics, we also assessed the impact of different workflows on phosphopeptide identification at varying false localization rate (FLR) thresholds (Figure 4B, C). After filtering with PSM FDR

0.01 and local FLR 0.01, the consensus workflow with quantms-rescoring features identified 20% and 23% more phosphorylated peptides than its counterpart without quantms-rescoring and single search engine. Notably, after filtering with PSM FDR 0.01, FLR 0.01 and protein FDR 0.01, 1,809 phosphorylated peptides were uniquely quantified by the quantms-rescoring enabled workflow compared to without quantms-rescoring. When considering -98 Da neutral losses, 2,191 phosphorylated peptides were uniquely quantified compared to without quantms-rescoring. At the phosphosite level, workflow with quantms-rescoring also uncovered 842 novel protein phosphorylation sites not reported in the other workflow configuration without quantms-rescoring (Figure 4D), and an additional 495 novel protein phosphorylation sites are reported when considering -98 Da neutral losses. Collectively, these results underscore the value of quantms-rescoring in enhancing both identification sensitivity and site-level localization in phosphoproteomics, establishing quantms as a robust solution for large-scale PTM data analysis.

4 Discussion

Advancements in deep learning-based tools have significantly improved the sensitivity of peptide-spectrum match (PSM) rescoring, yet their integration into streamlined, reproducible, and quantitative, scalable proteomics workflows has remained limited, especially for large-scale public data analysis. In this study, we demonstrate the systematic incorporation of quantms-rescoring into the open-source quantms pipeline and evaluate its impact across various experimental settings, including label-free quantification (LFQ), tandem mass tag (TMT)-based quantification, immunopeptidomics, and phosphoproteomics. Our results show that enhanced feature sets derived from MS²PIP and DeepLC not only increase the identification rates but also improve the quantification depth.

Importantly, we provide evidence that the improved identification process via rescoring propagates downstream into quantification, leading to increased numbers of proteins with re-

liable abundance estimates, as well as enhanced detection of differentially expressed proteins, and enhanced localization of PTMs.

To demonstrate the applicability and advantages of quantms integrated with quantms-rescoring in quantitative proteomics, we performed a series of reanalyses using publicly available large-scale datasets with well-established benchmarks. In label-free experiments, quantms-rescoring-enhanced workflows achieved significant increases in PSM and peptide identification rates while maintaining or improving quantification reproducibility across replicates. For TMT experiments, the integration of quantms-rescoring into quantms increased the number of quantifiable proteins. In more challenging applications, such as immunopeptidomics and phosphoproteomics, the deep learning-based rescoring pipeline consistently yielded higher identification rates and improved sensitivity. These improvements will contribute to meaningful biological insights, such as phosphosite discovery and the identification of differentially expressed proteins with potential clinical relevance. In addition, we counted the run times with different configurations and the added sensitivity from rescoring does add 15%-50% of run time, as shown as Supplementary Table ???. This can be alleviated by increasing the number of quantms-rescoring parallel processes.

Overall, the results highlight the synergistic benefits of integrating machine learning-based features within end-to-end, cloud-based proteomics workflows. By embedding quantms-rescoring into the quantms pipeline, we provide a practical and accessible solution for the community to leverage cutting-edge spectral prediction and retention time modeling without the need for complex manual configuration. This integrated approach enables reproducible reanalysis of large public datasets, increases identification confidence, and strengthens quantitative conclusions across diverse experimental designs. Future work will focus on expanding this approach to other types of post-translational modifications and exploring the integration of additional machine learning tools to further enhance proteomics data analysis capabilities.

Data availability

All datasets analyzed in this study are publicly available. Raw data and metadata can be obtained from the PRIDE Archive under the accessions PXD001819 (LFQ benchmark), PXD019643 (immunopeptidomics), and PXD026824 (phosphoproteomics), and from the CP-TAC data portal under PDC000127 (TMT). Sample and Data Relationship Files (SDRFs) used in this work are available in the project repository at <https://github.com/bigbio/quantms-rescori>. Where applicable, additional configuration files and parameters are included alongside the SDRFs to facilitate reuse.

Acknowledgments

We thank all those who supported this research, including funding bodies and the proteomics community for making proteomics data sets publicly available. Yasset Perez-Riverol would like to acknowledge funding from EMBL core funding, Wellcome grants (208391/Z/17/Z, 223745/Z/21/Z) and the BBSRC grant ‘DIA-Exchange’ [BB/X001911/1]. R.G., L.M., and R.B. acknowledge funding from [12B7123N,G010023N,G028821N,12A6L24N]. L.M. acknowledge funding from the Horizon Europe Projects BAXERNA 2.0 [101080544] and COMBINE [101191739], and from the Ghent University Concerted Research Action [BOF21/GOA/033]. L.M. is further supported by the CHIST-ERA project ODEEP-EU [G0GDEV23N].

References

- [1] Yasset Perez-Riverol, Chakradhar Bandla, Deepti J Kundu, Selvakumar Kamatchinathan, Jingwen Bai, Suresh Hewapathirana, Nithu Sara John, Ananth Prakash, Mathias Walzer, Shengbo Wang, and Juan Antonio Vizcaíno. The PRIDE database at 20 years: 2025 update. *Nucleic Acids Research*, 53(D1):D543–D553, January 2025.
- [2] Chengxin Dai, Julianus Pfeuffer, Hong Wang, Ping Zheng, Lukas Käll, Timo Sach-

- senberg, Vadim Demichev, Mingze Bai, Oliver Kohlbacher, and Yasset Perez-Riverol. quantms: a cloud-based pipeline for quantitative proteomics enables the reanalysis of public proteomics data. *Nature Methods*, July 2024.
- [3] Michael R Lazear. Sage: an open-source tool for fast proteomics searching and quantification at scale. *Journal of Proteome Research*, 22(11):3652–3659, 2023.
- [4] Sangtae Kim and Pavel A Pevzner. MS-GF+ makes progress towards a universal database search tool for proteomics. *Nat. Commun.*, 5(1):5277, October 2014.
- [5] Jimmy K Eng, Tahmina A Jahan, and Michael R Hoopmann. Comet: an open-source MS/MS sequence database search tool. *Proteomics*, 13(1):22–24, January 2013.
- [6] Lukas Käll, Jesse D Canterbury, Jason Weston, William Stafford Noble, and Michael J MacCoss. Semi-supervised learning for peptide identification from shotgun proteomics datasets. *Nature methods*, 4(11):923–925, 2007.
- [7] Jürgen Cox and Matthias Mann. MaxQuant enables high peptide identification rates, individualized p.p.b.-range mass accuracies and proteome-wide protein quantification. *Nature Biotechnology*, 26(12):1367–1372, December 2008.
- [8] Marc Vaudel, Julia M Burkhart, René P Zahedi, Eystein Oveland, Frode S Berven, Albert Sickmann, Lennart Martens, and Harald Barsnes. Peptideshaker enables reanalysis of ms-derived proteomics data sets. *Nature biotechnology*, 33(1):22–24, 2015.
- [9] Le-heng Wang, De-Quan Li, Yan Fu, Hai-Peng Wang, Jing-Fen Zhang, Zuo-Fei Yuan, Rui-Xiang Sun, Rong Zeng, Si-Min He, and Wen Gao. pFind 2.0: a software package for peptide and protein identification via tandem mass spectrometry. *Rapid Communications in Mass Spectrometry*, 21(18):2985–2991, September 2007.
- [10] Paolo Di Tommaso, Maria Chatzou, Evan W Floden, Pablo Prieto Barja, Emilio

- Palumbo, and Cedric Notredame. Nextflow enables reproducible computational workflows. *Nature Biotechnology*, 35(4):316–319, April 2017.
- [11] Chengxin Dai, Anja Füllgrabe, Julianus Pfeuffer, Elizaveta M. Solovyeva, Jingwen Deng, Pablo Moreno, Selvakumar Kamatchinathan, Deepti Jaiswal Kundu, Nancy George, Silvie Fexova, Björn Grüning, Melanie Christine Föll, Johannes Griss, Marc Vaudel, Enrique Audain, Marie Locard-Paulet, Michael Turewicz, Martin Eisenacher, Julian Uszkoreit, Tim Van Den Bossche, Veit Schwämmle, Henry Webel, Stefan Schulze, David Bouyssié, Savita Jayaram, Vinay Kumar Duggineni, Patroklos Samaras, Mathias Wilhelm, Meena Choi, Mingxun Wang, Oliver Kohlbacher, Alvis Brazma, Irene Papatheodorou, Nuno Bandeira, Eric W. Deutsch, Juan Antonio Vizcaíno, Mingze Bai, Timo Sachsenberg, Lev I. Levitsky, and Yasset Perez-Riverol. A proteomics sample metadata representation for multiomics integration and big data analysis. *Nature Communications*, 12(1):5854, October 2021.
- [12] Lennart Martens, Matthew Chambers, Marc Sturm, Darren Kessner, Fredrik Levan-der, Jim Shofstahl, Wilfred H. Tang, Andreas Römpp, Steffen Neumann, Angel D. Pizarro, Luisa Montecchi-Palazzi, Natalie Tasman, Mike Coleman, Florian Reisinger, Puneet Souda, Henning Hermjakob, Pierre-Alain Binz, and Eric W. Deutsch. mzML—a Community Standard for Mass Spectrometry Data. *Molecular & Cellular Proteomics*, 10(1):R110.000133, January 2011.
- [13] Johannes Griss, Andrew R Jones, Timo Sachsenberg, Mathias Walzer, Laurent Gatto, Jürgen Hartler, Gerhard G Thallinger, Reza M Salek, Christoph Steinbeck, Nadin Neuhauser, et al. The mztab data exchange format: communicating mass-spectrometry-based proteomics and metabolomics experimental results to a wider audience. *Molecular & Cellular Proteomics*, 13(10):2765–2775, 2014.
- [14] Felipe da Veiga Leprevost, Björn A Grüning, Saulo Alves Aflitos, Hannes L Röst, Julian Uszkoreit, Harald Barsnes, Marc Vaudel, Pablo Moreno, Laurent Gatto, Jonas Weber,

- et al. Biocontainers: an open-source and community-driven framework for software standardization. *Bioinformatics*, 33(16):2580–2582, 2017.
- [15] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino Da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C 'T Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene Van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan Van Der Lei, Erik Van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao, and Barend Mons. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3(1):160018, March 2016.
- [16] Louise M. Buur, Arthur Declercq, Marina Strobl, Robbin Bouwmeester, Sven Degroeve, Lennart Martens, Viktoria Dorfer, and Ralf Gabriels. MS² Rescore 3.0 Is a Modular, Flexible, and User-Friendly Platform to Boost Peptide Identifications, as Showcased with MS Amanda 3.0. *Journal of Proteome Research*, 23(8):3200–3207, August 2024.
- [17] Kevin L Yang, Fengchao Yu, Guo Ci Teo, Kai Li, Vadim Demichev, Markus Ralser, and Alexey I Nesvizhskii. Msbooster: improving peptide identification rates using deep learning-based features. *Nature Communications*, 14(1):4539, 2023.
- [18] Arthur Declercq, Robbin Bouwmeester, Cristina Chiva, Eduard Sabidó, Aurélie Hirschler, Christine Carapito, Lennart Martens, Sven Degroeve, and Ralf Gabriels. Updated MS²PIP web server supports cutting-edge proteomics applications. *Nucleic Acids Research*, 51:W338–W342, 2023.

- [19] Wen-Feng Zeng, Xie-Xuan Zhou, Sander Willems, Constantin Ammar, Maria Wahle, Isabell Bludau, Eugenia Voytik, Maximillian T. Strauss, and Matthias Mann. AlphaPept-Deep: a modular deep learning framework to predict peptide properties for proteomics. *Nature Communications*, 13(1):7238, 2022.
- [20] Robbin Bouwmeester, Ralf Gabriels, Niels Hulstaert, Lennart Martens, and Sven Degroeve. DeepLC can predict retention times for peptides that carry as-yet unseen modifications. *Nature Methods*, 18(11):1363–1369, November 2021.
- [21] Arthur Declercq, Robbe Devreese, Jonas Scheid, Caroline Jachmann, Tim Van Den Bossche, Annica Preikschat, David Gomez-Zepeda, Jeewan Babu Rijal, Aurélie Hirschler, Jonathan R Krieger, Tharan Srikumar, George Rosenberger, Claudia Martelli, Dennis Trede, Christine Carapito, Stefan Tenzer, Juliane S Walz, Sven Degroeve, Robbin Bouwmeester, Lennart Martens, and Ralf Gabriels. TIMS² rescore: A data dependent acquisition-parallel accumulation and serial fragmentation-optimized data-driven rescoring pipeline based on MS² rescore. *J. Proteome Res.*, 24(3):1067–1076, 2025.
- [22] William E Fondrie and William S Noble. mokapot: Fast and flexible semisupervised learning for peptide detection. *Journal of Proteome Research*, 20(4):1966–1971, 2021.
- [23] Claire Ramus, Agnès Hovasse, Marlène Marcellin, Anne-Marie Hesse, Emmanuelle Mouton-Barbosa, David Bouyssié, Sebastian Vaca, Christine Carapito, Karima Chaoui, Christophe Bruley, Jérôme Garin, Sarah Cianférani, Myriam Ferro, Alain Van Dorssaeler, Odile Burlet-Schiltz, Christine Schaeffer, Yohann Couté, and Anne Gonzalez De Peredo. Spiked proteomic standard dataset for testing label-free quantitative software and statistical methods. *Data in Brief*, 6:286–294, 2016.
- [24] Ana Marcu, Leon Bichmann, Leon Kuchenbecker, Daniel Johannes Kowalewski, Lena Katharina Freudenmann, Linus Backert, Lena Mühlenbruch, András Szolek,

- Maren Lübke, Philipp Wagner, Tobias Engler, Sabine Matovina, Jian Wang, Mathias Hauri-Hohl, Roland Martin, Konstantina Kapolou, Juliane Sarah Walz, Julia Velz, Holger Moch, Luca Regli, Manuela Silginer, Michael Weller, Markus W. Löffler, Florian Erhard, Andreas Schlosser, Oliver Kohlbacher, Stefan Stevanović, Hans-Georg Rammensee, and Marian Christoph Neidert. HLA Ligand Atlas: a benign reference of HLA-presented peptides to improve T-cell-based cancer immunotherapy. *Journal for ImmunoTherapy of Cancer*, 9(4):e002071, April 2021.
- [25] Adi Jacob Berger, Elinor Gigi, Lana Kupersmidt, Zohar Meir, Nancy Gavert, Yaara Zwang, Amir Prior, Shlomit Gilad, Uzi Harush, Izhak Haviv, Salomon M. Stemmer, Galia Blum, Emmanuelle Merquiol, Mariya Mardamshina, Sivan Kaminski Strauss, Gilgi Friedlander, Jair Bar, Iris Kamer, Yitzhak Reizel, Tamar Geiger, Yitzhak Pilpel, Yishai Levin, Amos Tanay, Baruch Barzel, Hadas Reuveni, and Ravid Straussman. IRS1 phosphorylation underlies the non-stochastic probability of cancer cells to persist during EGFR inhibition therapy. *Nature Cancer*, 2(10):1055–1070, 2021.
- [26] David J. Clark, Saravana M. Dhanasekaran, Francesca Petralia, Jianbo Pan, Xiaoyu Song, Yingwei Hu, Felipe Da Veiga Leprevost, Boris Reva, Tung-Shing M. Lih, Hui-Yin Chang, Weiping Ma, Chen Huang, Christopher J. Ricketts, Lijun Chen, Azra Krek, Yize Li, Dmitry Rykunov, Qing Kay Li, Lin S. Chen, Umut Ozbek, Suhas Vasaikar, Yige Wu, Seungyeul Yoo, Shrabanti Chowdhury, Matthew A. Wyczalkowski, Jiayi Ji, Michael Schnaubelt, Andy Kong, Sunantha Sethuraman, Dmitry M. Avtonomov, Minghui Ao, Antonio Colaprico, Song Cao, Kyung-Cho Cho, Selim Kalayci, Shiyong Ma, Wenke Liu, Kelly Ruggles, Anna Calinawan, Zeynep H. Gümüş, Daniel Geiszler, Emily Kawaler, Guo Ci Teo, Bo Wen, Yuping Zhang, Sarah Keegan, Kai Li, Feng Chen, Nathan Edwards, Phillip M. Pierorazio, Xi Steven Chen, Christian P. Pavlovich, A. Ari Hakimi, Gabriel Brominski, James J. Hsieh, Andrzej Antczak, Tatiana Omelchenko, Jan Lubinski, Maciej Wiznerowicz, W. Marston Linehan, Christopher R. Kinsinger, Mathangi

Thiagarajan, Emily S. Boja, Mehdi Mesri, Tara Hiltke, Ana I. Robles, Henry Rodriguez, Jiang Qian, David Fenyő, Bing Zhang, Li Ding, Eric Schadt, Arul M. Chinnaiyan, Zhen Zhang, Gilbert S. Omenn, Marcin Cieslik, Daniel W. Chan, Alexey I. Nesvizhskii, Pei Wang, Hui Zhang, Abdul Samad Hashimi, Alexander R. Pico, Alla Karpova, Alyssa Charamut, Amanda G. Paulovich, Amy M. Perou, Anna Malovannaya, Annette Marrero-Oliveras, Anupriya Agarwal, Barbara Hindenach, Barbara Pruetz, Beom-Jun Kim, Brian J. Druker, Chelsea J. Newton, Chet Birger, Corbin D. Jones, Cristina Tognon, D.R. Mani, Dana R. Valley, Daniel C. Rohrer, Daniel C. Zhou, Darlene Tansil, David Chesla, David Heiman, David Wheeler, Donghui Tan, Doug Chan, Emek Demir, Ewa Malc, Francesmary Modugno, Gaddy Getz, Galen Hostetter, George D. Wilson, Gerald W. Hart, Heng Zhu, Hongwei Liu, Houston Culpepper, Hua Sun, Hua Zhou, Jacob Day, James Suh, Jasmine Huang, Jason McDermott, Jeffrey R. Whiteaker, Jeffrey W. Tyner, Jennifer Eschbacher, Jin Chen, John McGee, Jun Zhu, Karen A. Ketchum, Karin D. Rodland, Karl Clauser, Karna Robinson, Karsten Krug, Katherine A. Hoadley, Ki Sung Um, Kim Elburn, Kimberly Holloway, Liang-Bo Wang, Lili Blumenberg, Linda Hannick, Liqun Qi, Lori J. Sokoll, MacIntosh Cornwell, Marc Loriaux, Marcin J. Domagalski, Marina A. Gritsenko, Matthew Anderson, Matthew E. Monroe, Matthew J. Ellis, Maureen Dyer, Meenakshi Anurag, Meghan C. Burke, Melissa Borucki, Michael A. Gillette, Michael J. Birrer, Michael Lewis, Michael M. Ittmann, Michael Smith, Michael Vernon, Michelle Chaikin, Milan G. Chheda, Munziba Khan, Nancy Roche, Nathan J. Edwards, Negin Vatanian, Nicole Tignor, Noam Beckmann, Pamela Grady, Patricia Castro, Paul Piehowski, Peter B. McGarvey, Piotr Mieczkowski, Pushpa Hariharan, Qingsong Gao, Rajiv Dhir, Ramani Bhupendra Kothadia, Ratna R. Thangudu, Rebecca Montgomery, Reyka G. Jayasinghe, Richard D. Smith, Robert Edwards, Robert Zelt, Ross Bremner, Ruiyang Liu, Runyu Hong, Sailaja Mareedu, Samuel H. Payne, Sandra Cottingham, Sanford P. Markey, Scott D. Jewell, Shalin Patel, Shankha Satpathy, Shannon Richey, Sherri R. Davies, Shuang Cai, Simina M.

- Boca, Snehal Patil, Sohini Sengupta, Sonya Carter, Stacey Gabriel, Stefani N. Thomas, Stephanie De Young, Stephen E. Stein, Steven A. Carr, Steven M. Foltz, Sue Hilsenbeck, Tanya Krubit, Tao Liu, Tara Skelly, Thomas Westbrook, Uma Borate, Uma Velvulou, Vladislav A. Petyuk, William E. Bocik, Xi Chen, Yan Shi, Yifat Geffen, Yihao Lu, Ying Wang, Yosef Maruvka, Zhi Li, Zhiao Shi, and Zhidong Tu. Integrated proteogenomic characterization of clear cell renal cell carcinoma. *Cell*, 179(4):964–983.e31, 2019.
- [27] Robbin Bouwmeester, Alireza Nameni, Arthur Declercq, Robbe Devreese, Kevin Velghe, Vladimir Gorshkov, Pelayo A Penanes, Frank Kjeldsen, Magali Rompais, Christine Carapito, et al. Deeplc introduces transfer learning for accurate lc retention time prediction and adaptation to substantially different modifications and setups. *bioRxiv*, pages 2025–06, 2025.
- [28] Sven Nahnsen, Andreas Bertsch, Jörg Rahnenführer, Alfred Nordheim, and Oliver Kohlbacher. Probabilistic consensus scoring improves tandem mass spectrometry peptide identification. *Journal of Proteome Research*, 10(8):3332–3343, 2011.
- [29] Julianus Pfeuffer, Chris Bielow, Samuel Wein, Kyowon Jeong, Eugen Netz, Axel Walter, Oliver Alka, Lars Nilse, Pasquale Domenico Colaianni, Douglas McCloskey, et al. Openms 3 enables reproducible analysis of large-scale mass spectrometry data. *Nature methods*, 21(3):365–367, 2024.
- [30] Damian Fermin, Dmitry Avtonomov, Hyungwon Choi, and Alexey I. Nesvizhskii. LuciPHOr2: site localization of generic post-translational modifications from tandem mass spectrometry data. *Bioinformatics*, 31(7):1141–1143, 2015.
- [31] Wenjie Yang, Hualin Chen, Lin Ma, Jie Dong, Mengchao Wei, Xiaoqiang Xue, Yingjie Li, Zhaoheng Jin, Weifeng Xu, and Zhigang Ji. A comprehensive analysis of the FOX family for predicting kidney renal clear cell carcinoma prognosis and the oncogenic role of FOXG1. *Aging*, 14(24):10107–10124, December 2022.

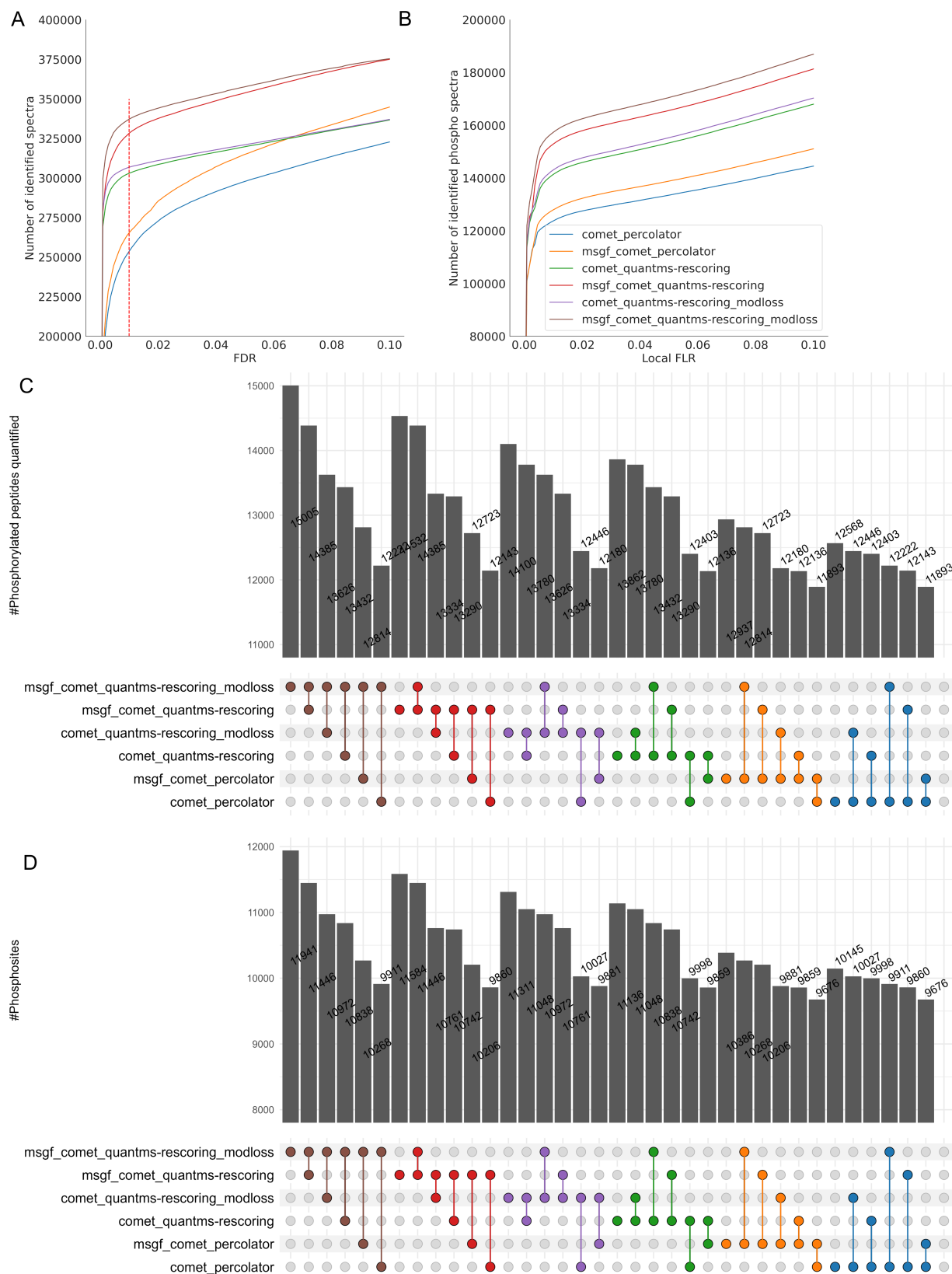


Figure 5: Comparison of phosphorylated peptide identification results for different workflow settings on PXD026824. (A) Line plot showing the number of spectra identified for four workflow settings at different PSM FDR levels. (B) Line plot showing the number of phosphorylated spectra identified for four workflow settings at different local false localization rate levels after an FDR of less than 0.01 at the PSM level. (C) Venn diagram of peptides quantified for four settings. (D) Venn diagram of protein phosphosites at protein FDR 0.01